#1 · Safety and Health of Welders

What is ANSI Z49.1?

Answer

The title is Safety in Welding, Cutting, and Allied Processes. ANSI is the American National Standards Institute, and the American Welding Society publishes it. Many shops and schools use it as a baseline safety rulebook.

#2 · Safety and Health of Welders

What does OSHA stand for and what does it do?

Answer

OSHA is the Occupational Safety and Health Administration. It is the US agency that sets workplace safety rules. If your school or employer is stricter, follow the stricter rule.

#3 · Safety and Health of Welders

What score is expected on the Level 1 SENSE safety exam?

Answer

Most module exams need 75 percent, but the Level 1 safety exam needs a perfect 100 percent. Check your program's rules, including how retakes work. One missed safety fact can mean a burn, a shock or a fire.

#4 · Safety and Health of Welders

Which three kinds of light does a welding arc give off?

Answer

An arc gives off ultraviolet (UV), visible and infrared (IR) light. UV burns skin and eyes without being seen. Infrared is felt as heat and can damage the eye over time.

#5 · Safety and Health of Welders

What is arc flash (welder's flash)?

Answer

It is a sunburn of the cornea, the clear front of the eye, and its medical name is photokeratitis. Symptoms usually begin several hours after exposure. The eyes feel gritty, water and become very sensitive to light.

#6 · Safety and Health of Welders

What is a shade number on a welding lens?

Answer

It rates how dark the filter lens is. A higher number means a darker lens. It is a safety rating chosen by process and current, not a comfort setting.

#7 · Safety and Health of Welders

What shade range is typical for stick (SMAW) welding?

Answer

Often shade 10 to 14, depending on electrode size and current. Always check the current Z49.1 table and your employer's policy.

#8 · Safety and Health of Welders

What shade range is typical for GMAW and FCAW (MIG and flux-cored)?

Answer

Often shade 10 to 14, depending on current. Higher amperage calls for a darker shade.

#9 · Safety and Health of Welders

What shade range is typical for GTAW (TIG)?

Answer

About shade 8 to 14, getting darker as amperage rises. Low-current TIG can use a lighter lens than stick welding.

#10 · Safety and Health of Welders

What shade range is typical for light oxyfuel welding?

Answer

About shade 4 to 6. Torch brazing is lighter still, about shade 3 or 4. Plasma arc cutting is roughly shade 9 to 12 depending on current.

#11 · Safety and Health of Welders

Why must safety glasses be worn under the welding helmet?

Answer

When the helmet is flipped up to chip slag or grind, flying particles can reach the eyes. Slag is the hard crust over a weld and hot chips can fly in any direction. Grinding also needs a full face shield over the glasses.

#12 · Safety and Health of Welders

What is the advantage and limit of an auto-darkening lens?

Answer

It is light until the arc starts and then darkens in a tiny fraction of a second. It must still be rated for the process and set to the right shade. The helmet is lowered before the arc starts.

#13 · Safety and Health of Welders

Why are synthetic fabrics such as nylon and polyester banned in the weld shop?

Answer

They melt when hit by sparks or spatter. Melted fabric sticks to skin and makes burns much worse. Flame-resistant cotton, wool and leather are the common choices.

#14 · Safety and Health of Welders

Why should welding pants have no cuffs?

Answer

A cuff catches a hot spark like a pocket does. Pants should also cover the tops of high-top leather boots. Pocket flaps should be closed or pockets buttoned.

#15 · Safety and Health of Welders

What extra protection is worn for overhead welding?

Answer

A cap, a leather cape or jacket and ear protection. A hot piece of metal can fall into the ear canal. Overhead work throws sparks onto the head and shoulders.

#16 · Safety and Health of Welders

Why must welding gloves be dry and in good condition?

Answer

A wet or torn glove gives little protection from electric shock or heat. Gloves are made of leather. Replace worn ones.

#17 · Safety and Health of Welders

What is open circuit voltage?

Answer

It is the voltage at the electrode when no arc is struck. For many stick machines it is commonly about 60 to 80 volts. That is lower than a wall outlet but can still be dangerous.

#18 · Safety and Health of Welders

What does Ohm's law say, and how does it explain shock danger?

Answer

Current equals voltage divided by resistance, I = V ÷ R. Wet skin has much lower resistance than dry skin, so far more current flows. At 80 V, roughly 1,000 Ω gives 80 mA, while roughly 100,000 Ω gives 0.8 mA.

#19 · Safety and Health of Welders

About how much body current can make a person unable to let go?

Answer

Around 10 to 20 milliamps can clamp the muscles. Currents around 100 milliamps and above can cause a deadly heart rhythm. These are general figures and wet conditions make them much worse.

#20 · Safety and Health of Welders

Where should the work lead (ground clamp) be attached?

Answer

On the work or the table close to where you are welding. This keeps welding current from traveling through chains, cranes or building steel. Unplanned paths can cause shock and fire.

#21 · Safety and Health of Welders

What should a welder do about damaged cables or holders?

Answer

Inspect cables, electrode holders and plugs before use and replace any with cracked or bare insulation. Only qualified people service welding machines. Shut off and unplug a machine before it is opened.

#22 · Safety and Health of Welders

What is the first rule for avoiding welding fume?

Answer

Keep your head out of the fume plume, the rising column of smoke above the weld. A small shift of your head greatly reduces fume you breathe. Position the work so the plume rises away from your face.

#23 · Safety and Health of Welders

What is the difference between general and local exhaust ventilation?

Answer

General ventilation moves air through the whole room with fans and open doors. Local exhaust captures fume at the source with a hood or extractor nozzle near the arc. Local exhaust is usually far more effective.

#24 · Safety and Health of Welders

Why should oxygen never be used to freshen the air?

Answer

Oxygen makes fires burn violently. Fresh air is the correct ventilation gas. This rule also applies inside confined spaces.

#25 · Safety and Health of Welders

What hazard comes from welding galvanized steel?

Answer

Burning zinc makes zinc oxide fume, which causes metal fume fever. It feels like the flu, with fever, chills, nausea and a metallic taste, usually hours after exposure. Grind off the coating from the weld area when you can and ventilate well.

#26 · Safety and Health of Welders

Why is welding stainless steel a fume hazard?

Answer

It contains chromium, and welding can make hexavalent chromium, linked to serious lung harm and cancer. The exposure limit is very low. Use local exhaust and the correct respirator.

#27 · Safety and Health of Welders

Why is cadmium a concern in welding and brazing?

Answer

Cadmium, found in old plating and some silver brazing alloys, is extremely toxic when heated. Stop work and ask for help if you suspect it. Lead in old paint and beryllium in some copper alloys are also very hazardous.

#28 · Safety and Health of Welders

Why must chlorinated solvents be kept away from welding?

Answer

Arc light can break their vapors down into phosgene, a deadly gas. Never weld where vapors from degreasers or cleaners are present. Let cleaned parts dry fully.

#29 · Safety and Health of Welders

What is hot work?

Answer

Any work that makes sparks, flame or heat, including welding, cutting, brazing and grinding. Many workplaces require a written hot work permit before it starts. The permit confirms inspection and fire protection.

#30 · Safety and Health of Welders

How far must combustibles be kept from hot work?

Answer

At least 35 feet, about 11 meters. If they cannot be moved, cover them with approved fire-resistant covers or shield them with welding blankets and screens.

#31 · Safety and Health of Welders

What is the job of a fire watch?

Answer

A fire watch is a person whose only job is to watch for fires. The watch stays during the work and for at least 30 minutes after it ends. Some employers or permits require longer, so check.

#32 · Safety and Health of Welders

Welding stops at 2:15 PM. Until when must the fire watch stay at minimum?

Answer

Until at least 2:45 PM, because the minimum is 30 minutes after the work ends. A longer time may be required by the permit or employer. Smoldering material can ignite after the sparks stop.

#33 · Safety and Health of Welders

Why must a drum that held fuel never be welded until it is made safe?

Answer

It must first be properly cleaned, tested and made safe. Leftover flammable vapor can explode, and even an empty drum is dangerous. Never assume a container is empty.

#34 · Safety and Health of Welders

What does PASS stand for with a fire extinguisher?

Answer

Pull the pin, Aim at the base of the fire, Squeeze the handle, Sweep side to side. Use an extinguisher only on a small fire that is safe to fight. If it grows, leave and raise the alarm.

#35 · Safety and Health of Welders

How must compressed gas cylinders be secured?

Answer

Upright, with a chain or strap, whether in use, in storage or in transport. Keep the valve protection cap on when not in use and when moving. A full cylinder holds roughly 2,200 psi, so a broken valve is a serious danger.

#36 · Safety and Health of Welders

What should you do before attaching a regulator to a cylinder?

Answer

Stand to the side of the valve outlet and crack the valve, which means open it briefly, to blow out dirt, then close it. Open the valve slowly afterward. Never stand in front of the outlet.

#37 · Safety and Health of Welders

How far open should oxygen and acetylene cylinder valves be?

Answer

Open an oxygen valve fully so the back seal is set, as the supplier directs. Open an acetylene valve only about one and a half turns and leave the wrench on the valve. That allows a fast shutoff.

#38 · Safety and Health of Welders

What is the maximum safe acetylene working pressure?

Answer

15 psi gauge. Above that, acetylene becomes unstable. Also keep acetylene cylinders upright.

#39 · Safety and Health of Welders

How must oxygen cylinders be stored relative to fuel gas?

Answer

At least 20 feet away from fuel gas cylinders, or separated by a noncombustible barrier at least 5 feet high with a fire rating of at least one half hour. Never use oil or grease on oxygen equipment, since oil and pure oxygen can ignite on contact.

#40 · Safety and Health of Welders

What is the safe oxygen range for working in a confined space?

Answer

About 19.5 to 23.5 percent, with normal air near 20.9 percent. Below 19.5 percent is oxygen-deficient. Above 23.5 percent is oxygen-enriched and fires burn much faster.

#41 · Safety and Health of Welders

Why are argon and carbon dioxide dangerous in a pit or tank?

Answer

They are heavier than air, settle low and displace oxygen. In a 10 ft × 10 ft × 8 ft room (800 ft^3), 100 ft^3 of argon leaking in could drop oxygen to about 18.3 percent. That is below the safe 19.5 percent minimum.

#42 · Safety and Health of Welders

What are the main dangers of a confined space?

Answer

Bad air, fire and explosion, and difficulty of rescue. A permit-required confined space needs a written entry permit. Test the air before entry and keep monitoring while working.

#43 · Safety and Health of Welders

What are the OSHA noise action level and permissible exposure limit?

Answer

The action level is 85 dBA averaged over 8 hours, and the permissible limit is 90 dBA for an 8-hour day. Every 5 dB above 90 halves the allowed time. At 100 dBA the limit is 2 hours.

#44 · Safety and Health of Welders

How long is exposure allowed at 95 dBA?

Answer

4 hours. The formula is time = 8 ÷ 2^((L – 90) ÷ 5), so 8 ÷ 2^1 = 4. Each extra 5 dB halves the time again.

#45 · Safety and Health of Welders

What are good ergonomic habits for welders?

Answer

Set the work at a comfortable height, change positions often, keep wrists straight and lift with the legs. Use a cart or crane for heavy items. Support heavy cables on a hook so they do not drag your arm.

#46 · Safety and Health of Welders

What are the three main tools of OSHA's Hazard Communication Standard?

Answer

Container labels, Safety Data Sheets (SDS) and worker training. The standard follows GHS, the Globally Harmonized System. Workers have a right to know about the chemicals they use.

#47 · Safety and Health of Welders

What does a hazard label show, and what are the two signal words?

Answer

A product name, a signal word, pictograms and hazard statements. The signal word is Danger for more severe hazards and Warning for less severe ones. A pictogram is a red-bordered diamond with a symbol such as a flame.

#48 · Safety and Health of Welders

What is a Safety Data Sheet?

Answer

A document with 16 sections that covers hazards, first aid, fire fighting, handling and storage, exposure limits and protective equipment. Welding products such as electrodes and fluxes have one. It must be available to workers during their shift.

#49 · Safety and Health of Welders

What are the steps of lockout and tagout under 29 CFR 1910.147?

Answer

Notify affected workers, shut the machine down, isolate every energy source, apply your own lock and tag, release stored energy, then verify by trying to start it. Each worker uses a personal lock that only that worker removes. A lock is stronger than a tag alone.

#50 · Safety and Health of Welders

What is the first step when someone is being shocked by a live source?

Answer

Do not touch the person while they may still be in contact. Turn off the power first, or move them with a dry, non-conductive object. Then call for emergency help and begin CPR if trained and needed.

#51 · Drawings and Welding Symbols

What is a print or drawing for a welder?

Answer

It is a set of instructions in a shared picture language. It tells you what to make, how big, which joints to weld and how each weld should look.

#52 · Drawings and Welding Symbols

What kind of projection do US drawings use?

Answer

US drawings use third angle projection. The top view sits above the front view and the right side view sits to the right of it.

#53 · Drawings and Welding Symbols

Where is the top view placed on a US print?

Answer

The top view is placed directly above the front view. This lets features line up from one view to the next.

#54 · Drawings and Welding Symbols

Where is the right side view placed on a US print?

Answer

The right side view is placed to the right of the front view. It lines up with the front view across the sheet.

#55 · Drawings and Welding Symbols

What does an auxiliary view show?

Answer

It looks straight at a sloped face of the part. This shows that face in its true shape and size.

#56 · Drawings and Welding Symbols

What does a section view show?

Answer

It shows the part as if it were cut open. The cut surface is shaded with thin slanted section lines.

#57 · Drawings and Welding Symbols

What does a cutting plane line show?

Answer

It shows where a section view was cut. Arrows on its ends show the direction you look.

#58 · Drawings and Welding Symbols

What is an object line?

Answer

An object line is a thick, solid line. It shows the visible edges and outline of the part.

#59 · Drawings and Welding Symbols

What is a hidden line?

Answer

A hidden line is a medium line made of short dashes. It shows edges behind a surface that cannot be seen from that view.

#60 · Drawings and Welding Symbols

What is a center line?

Answer

A center line is a thin line of long dash, short dash, long dash. It marks the center of holes, circles and round parts.

#61 · Drawings and Welding Symbols

What is the job of a dimension line?

Answer

A dimension line is a thin line with an arrowhead at each end. It carries the number that gives the distance.

#62 · Drawings and Welding Symbols

What is an extension line?

Answer

An extension line is a thin line that reaches out from the part. It lets the dimension line sit clear of the picture.

#63 · Drawings and Welding Symbols

What is a leader line?

Answer

A leader line is a thin line with an arrowhead that points from a note to a feature. The welding symbol sits on a leader that ends in an arrow.

#64 · Drawings and Welding Symbols

What does a break line show?

Answer

A break line is wavy or zigzag. It shows that a long part was shortened on the page to save room.

#65 · Drawings and Welding Symbols

What is the title block?

Answer

It is the box in a corner of the sheet. It holds the part name, drawing number, scale, material, date and general tolerances.

#66 · Drawings and Welding Symbols

Should you measure a print with a ruler to find a size?

Answer

No. The print may have been copied or shrunk, so read the printed dimension. If two numbers disagree, ask before cutting metal.

#67 · Drawings and Welding Symbols

What is a tolerance?

Answer

A tolerance is the total amount a size is allowed to vary. It tells you how much error is acceptable.

#68 · Drawings and Welding Symbols

What is a bilateral tolerance?

Answer

The size may vary both above and below the given number, such as 4.000 ± 0.010. That part is good from 3.990 to 4.010.

#69 · Drawings and Welding Symbols

What is a unilateral tolerance?

Answer

The size may vary in only one direction, such as 4.000 plus 0.020 and minus 0.000. That part is good from 4.000 to 4.020.

#70 · Drawings and Welding Symbols

What are limit dimensions?

Answer

The print gives the smallest and largest allowed sizes directly, such as 3.990 and 4.010. Any size between them is good.

#71 · Drawings and Welding Symbols

What tolerance applies when none is written beside a dimension?

Answer

The general tolerance in the title block applies. It often depends on the number of decimal places in the dimension.

#72 · Drawings and Welding Symbols

What does scale mean on a drawing?

Answer

Scale is the ratio of drawing size to real size. A scale of 1:1 is full size and 1:2 is half size.

#73 · Drawings and Welding Symbols

A line measures 3 inches on a drawing at 1:4 scale. What is the real length?

Answer

Real length is 3 × 4 = 12 inches. At 1:4 the real part is four times the drawn size.

#74 · Drawings and Welding Symbols

What does a 2:1 scale mean?

Answer

The drawing is twice the real size. It is used for small parts.

#75 · Drawings and Welding Symbols

What is a bill of materials?

Answer

A bill of materials is a list of every piece in an assembly. It gives item number, quantity, description, size and material.

#76 · Drawings and Welding Symbols

What does the item number in a bill of materials connect to?

Answer

It matches a circled number or balloon on the drawing. That balloon points to the piece on the print.

#77 · Drawings and Welding Symbols

What are the five basic joint types?

Answer

The five are butt, corner, tee, lap and edge. A joint is where pieces meet, and a weld is what goes in it.

#78 · Drawings and Welding Symbols

What is a tee joint?

Answer

One piece stands at about a right angle against the face of another piece. The two pieces form a T.

#79 · Drawings and Welding Symbols

What is a lap joint?

Answer

Two pieces overlap each other in parallel planes. The weld is made along the overlapping edges.

#80 · Drawings and Welding Symbols

What is the root of a weld?

Answer

The root is the point where the back of the weld meets the base metal surfaces. In a fillet weld it is the deepest point.

#81 · Drawings and Welding Symbols

What is the toe of a weld?

Answer

The toe is where the weld face meets the base metal. A fillet weld has two toes.

#82 · Drawings and Welding Symbols

What is the leg of a fillet weld?

Answer

The leg is the distance from the root to the toe, measured along the base metal. A fillet weld has two legs.

#83 · Drawings and Welding Symbols

What is the throat of a fillet weld?

Answer

The throat is the shortest distance from the root through the weld to its face. For an equal leg flat fillet it is about 0.707 times the leg size.

#84 · Drawings and Welding Symbols

What is the effective throat of a 1/4 inch equal leg fillet?

Answer

About 0.250 × 0.707 = 0.177 inch. The effective throat is about 0.707 times the leg size.

#85 · Drawings and Welding Symbols

What is the bevel angle?

Answer

The bevel angle is the angle between the bevel face and a line square to the surface. It measures the cut on one piece.

#86 · Drawings and Welding Symbols

What is the groove angle?

Answer

The groove angle, or included angle, is the total angle between the groove faces of both pieces. In a symmetrical V it is two times the bevel angle.

#87 · Drawings and Welding Symbols

Both plates in a V groove are beveled 30°. What is the groove angle?

Answer

The groove angle is 30° + 30° = 60°. It is the total included angle.

#88 · Drawings and Welding Symbols

What is the root face?

Answer

The root face, or land, is the flat unbeveled part of the groove face at the root. It helps stop burn through.

#89 · Drawings and Welding Symbols

What is the root opening?

Answer

The root opening is the gap between the pieces at the root of the joint. A larger gap helps penetration but needs more filler metal.

#90 · Drawings and Welding Symbols

In a welding symbol, what is the reference line?

Answer

The reference line is the horizontal line. All other symbol parts are placed around it in set positions.

#91 · Drawings and Welding Symbols

What does the arrow of a welding symbol touch?

Answer

It touches the joint to be welded. The side it touches is the arrow side and the opposite side is the other side.

#92 · Drawings and Welding Symbols

Where does a weld symbol go to show a weld on the arrow side?

Answer

It goes below the reference line. A symbol above the line means the other side.

#93 · Drawings and Welding Symbols

What does the tail of a welding symbol hold?

Answer

The tail is a fork at the end of the reference line. It holds notes such as the process, specification or WPS number, and is left off if not needed.

#94 · Drawings and Welding Symbols

Where is the size of a fillet weld written on the symbol?

Answer

The size is written to the left of the fillet symbol. The vertical side of the triangle is also always on the left.

#95 · Drawings and Welding Symbols

What does a fillet symbol with no length mean?

Answer

The weld runs the full length of the joint. It stops at a sudden change in the joint direction or at the end.

#96 · Drawings and Welding Symbols

What do the numbers 3-10 to the right of a fillet symbol mean?

Answer

Each weld is 3 inches long, and the welds are 10 inches apart center to center. The first number is length and the second is pitch.

#97 · Drawings and Welding Symbols

What does a circle at the arrow and reference line junction mean?

Answer

It is the weld all around symbol. The weld goes completely around the joint, such as around a pipe or tube.

#98 · Drawings and Welding Symbols

What does a flag at the arrow and reference line junction mean?

Answer

It is the field weld symbol, meaning the weld is made at the job site rather than in the shop. The flag points toward the tail.

#99 · Drawings and Welding Symbols

How do the backing and melt through symbols differ?

Answer

Both are half circles on the side opposite the groove weld symbol. An open half circle means backing, and a filled half circle means melt through.

#100 · Drawings and Welding Symbols

What do the letters G, M and C beside a contour line mean?

Answer

They name the finish method: G is grinding, M is machining and C is chipping. The contour line itself shows flush, convex or concave.

#101 · Level 1 Arc Welding Processes

What does SMAW stand for, and what is it commonly called?

Answer

Shielded metal arc welding. It is commonly called stick welding because it uses a flux-covered metal rod as the electrode.

#102 · Level 1 Arc Welding Processes

What does GMAW stand for, and what is it commonly called?

Answer

Gas metal arc welding. It is commonly called MIG and feeds a continuous solid wire while shielding gas flows around the arc.

#103 · Level 1 Arc Welding Processes

What does GTAW stand for, and what is it commonly called?

Answer

Gas tungsten arc welding. It is commonly called TIG and uses a non-consumable tungsten electrode.

#104 · Level 1 Arc Welding Processes

Why must molten weld metal be shielded from the air?

Answer

Oxygen and nitrogen in the air dissolve into molten steel. They cause gas pockets and brittle metal in the weld.

#105 · Level 1 Arc Welding Processes

What is a constant current (CC) power source?

Answer

A machine that holds amperage nearly steady even when arc length changes a little. It is also called a drooping characteristic machine and is used for SMAW and GTAW.

#106 · Level 1 Arc Welding Processes

What is a constant voltage (CV) power source?

Answer

A machine that holds voltage nearly steady and lets current change. It is also called a flat characteristic machine and is used for GMAW and FCAW.

#107 · Level 1 Arc Welding Processes

What is open circuit voltage (OCV)?

Answer

The voltage across the machine terminals when no arc is burning. For stick machines it is commonly 60 to 80 volts.

#108 · Level 1 Arc Welding Processes

What is duty cycle?

Answer

The part of a ten-minute period that a machine can weld at its rated output without overheating. A machine rated 300 amps at 60 percent can weld 6 minutes out of every 10.

#109 · Level 1 Arc Welding Processes

What is DCEP?

Answer

Direct current electrode positive, also called reverse polarity (DCRP). The electrode lead is on the positive terminal and penetration is deep.

#110 · Level 1 Arc Welding Processes

What is DCEN?

Answer

Direct current electrode negative, also called straight polarity (DCSP). The electrode lead is on the negative terminal, the electrode melts faster and penetration is shallower.

#111 · Level 1 Arc Welding Processes

How does AC share heat between electrode and work?

Answer

About evenly, giving medium penetration. AC reverses direction 120 times each second on a 60 hertz supply.

#112 · Level 1 Arc Welding Processes

What is arc blow?

Answer

The sideways push of an arc caused by magnetic fields. Moving the ground clamp, changing weld direction or using AC can reduce it.

#113 · Level 1 Arc Welding Processes

What are the three jobs of an SMAW electrode coating?

Answer

It makes a shielding gas around the pool, forms a slag layer over the cooling weld, and adds alloying elements and arc stabilizers.

#114 · Level 1 Arc Welding Processes

What is slag, and why is it removed?

Answer

A layer formed by the burned flux coating that protects the cooling weld. It is chipped off between passes because slag left behind causes slag inclusions.

#115 · Level 1 Arc Welding Processes

What is the work clamp for?

Answer

It connects the workpiece or table to the machine so the welding circuit is complete. It is also called the ground clamp.

#116 · Level 1 Arc Welding Processes

In E7018, what does the E mean?

Answer

Electrode. It is the first letter of every AWS stick electrode classification.

#117 · Level 1 Arc Welding Processes

In E7018, what do the first two digits mean?

Answer

Minimum tensile strength in thousands of pounds per square inch. 70 means 70,000 psi.

#118 · Level 1 Arc Welding Processes

In E7018, what does the third digit 1 mean?

Answer

The electrode can be used in all positions: flat, horizontal, vertical and overhead.

#119 · Level 1 Arc Welding Processes

What is the fourth digit in a stick electrode number?

Answer

It is read with the third digit and tells the coating type and the current the rod runs on.

#120 · Level 1 Arc Welding Processes

What kind of coating does E6010 have?

Answer

A cellulosic coating. It gives a forceful, fast-freezing arc with deep penetration and thin slag.

#121 · Level 1 Arc Welding Processes

What is E6010 classically used for?

Answer

The root pass on pipe. It digs deep and can weld through light rust and paint.

#122 · Level 1 Arc Welding Processes

What is E6013 known for?

Answer

A soft, easy arc with light penetration and heavy slag. It is common for sheet metal and general shop work.

#123 · Level 1 Arc Welding Processes

What type of electrode is E7018?

Answer

A low-hydrogen, iron powder electrode. It makes smooth, crack-resistant welds with slag that peels easily.

#124 · Level 1 Arc Welding Processes

Which currents does E7018 run on?

Answer

DCEP or AC. Confirm polarity on the electrode package.

#125 · Level 1 Arc Welding Processes

When is E7018 the standard choice?

Answer

For structural steel and for thicker or restrained joints, because it is strong and crack resistant.

#126 · Level 1 Arc Welding Processes

What is E7024 used for?

Answer

High deposition welding in the flat position and horizontal fillets only. It has a heavy iron powder coating.

#127 · Level 1 Arc Welding Processes

Why is hydrogen a problem in a weld?

Answer

Hydrogen from moisture can dissolve in the weld and cause cracks that may appear hours or days after welding.

#128 · Level 1 Arc Welding Processes

Why does E7018 come in a hermetically sealed container?

Answer

Low-hydrogen coatings absorb moisture from humid air. The sealed container keeps the coating dry until opened.

#129 · Level 1 Arc Welding Processes

What is a rod oven (holding oven) for?

Answer

Storing opened low-hydrogen electrodes to keep them dry, commonly around 250°F to 300°F. The exact setting comes from the code, WPS and manufacturer.

#130 · Level 1 Arc Welding Processes

What is a good general arc length in SMAW?

Answer

About equal to the diameter of the core wire. For a 1/8 inch rod that is about 1/8 inch.

#131 · Level 1 Arc Welding Processes

Why does E7018 need an especially short arc?

Answer

A long arc lets air into the pool and causes porosity.

#132 · Level 1 Arc Welding Processes

Roughly what amperage range is typical for 1/8 inch E7018?

Answer

Often between 90 and 140 amps. The range printed on the box is the best guide.

#133 · Level 1 Arc Welding Processes

What is travel angle?

Answer

The tilt of the electrode in the direction of travel, usually 5° to 15° in stick welding.

#134 · Level 1 Arc Welding Processes

What is work angle?

Answer

The tilt of the electrode relative to the sides of the joint. It is about 90° on a flat butt weld and about 45° on a T-joint fillet.

#135 · Level 1 Arc Welding Processes

What is the usual SMAW rod motion, drag or push?

Answer

Drag, with the tip leading the pool. Pushing the rod tends to trap slag.

#136 · Level 1 Arc Welding Processes

Why fill the crater at the end of a weld?

Answer

An unfilled crater can crack. Pause briefly at the end to fill it.

#137 · Level 1 Arc Welding Processes

What parts make up a GMAW setup?

Answer

A CV power source, wire feeder with drive rolls, gun cable with a liner, contact tip, nozzle, and a gas cylinder with regulator and flowmeter.

#138 · Level 1 Arc Welding Processes

What does the contact tip do?

Answer

It passes welding current to the wire as the wire leaves the gun.

#139 · Level 1 Arc Welding Processes

In ER70S-6, what does ER mean?

Answer

The product can be used as either an electrode or a rod.

#140 · Level 1 Arc Welding Processes

In ER70S-6, what does S mean?

Answer

Solid wire. It is a solid electrode, not a flux cored one.

#141 · Level 1 Arc Welding Processes

In ER70S-6, what does the 70 mean?

Answer

A minimum tensile strength of 70 ksi, or 70,000 psi.

#142 · Level 1 Arc Welding Processes

What does the 6 in ER70S-6 tell you?

Answer

The chemical composition. The -6 wire has extra manganese and silicon, which are deoxidizers.

#143 · Level 1 Arc Welding Processes

What is a deoxidizer in a welding wire?

Answer

An element, such as manganese or silicon, that binds with oxygen so it does not cause gas pockets or weak metal.

#144 · Level 1 Arc Welding Processes

What are the properties of 100 percent CO2 as a GMAW gas?

Answer

It is the cheapest gas and gives deep penetration, but the arc is rougher and spatter is higher.

#145 · Level 1 Arc Welding Processes

What is C25 shielding gas?

Answer

A mixture of 75 percent argon and 25 percent CO2. It is very common for mild steel and balances arc quality, spatter and penetration.

#146 · Level 1 Arc Welding Processes

What gas is needed for spray transfer?

Answer

A mostly argon gas, commonly 90 percent argon or richer, such as 90/10 argon and CO2.

#147 · Level 1 Arc Welding Processes

What is a typical GMAW shielding gas flow rate?

Answer

About 20 to 35 cubic feet per hour (CFH). Check the procedure.

#148 · Level 1 Arc Welding Processes

What can happen when GMAW gas flow is too high?

Answer

Turbulence can pull air into the shield, and gas is wasted.

#149 · Level 1 Arc Welding Processes

Why is GMAW poor outdoors?

Answer

Wind blows the shielding gas away from the pool, so porosity results unless wind screens are used.

#150 · Level 1 Arc Welding Processes

What is metal transfer?

Answer

How melted wire crosses the arc to the weld pool.

#151 · Level 1 Arc Welding Processes

What is short circuit transfer?

Answer

The wire tip touches the pool, shorts out and a drop is pinched off, many times each second. It uses low voltage and current.

#152 · Level 1 Arc Welding Processes

When is short circuit transfer used?

Answer

On thin metal, in out-of-position welding and for root passes, because the pool stays small and cool.

#153 · Level 1 Arc Welding Processes

What is the main defect risk of short circuit transfer?

Answer

Cold lap, meaning lack of fusion with the base metal, because heat input is low.

#154 · Level 1 Arc Welding Processes

What is globular transfer?

Answer

Large drops, bigger than the wire diameter, fall across the arc irregularly. It spatters a lot and is used in the flat position only.

#155 · Level 1 Arc Welding Processes

Which gas tends to cause globular transfer even at high settings?

Answer

100 percent CO2.

#156 · Level 1 Arc Welding Processes

What is spray transfer?

Answer

A fine stream of tiny droplets crosses the arc with little spatter and deep penetration. It needs high voltage and current and a mostly argon gas.

#157 · Level 1 Arc Welding Processes

What is pulsed spray transfer?

Answer

Current switches between a high peak and a low background. It gives spray droplets with less heat, so it works on thinner metal and more positions.

#158 · Level 1 Arc Welding Processes

In GMAW, what does wire feed speed control?

Answer

Amperage. Faster feed means more current because more wire must be melted.

#159 · Level 1 Arc Welding Processes

In GMAW, what does voltage control?

Answer

Arc length and bead width. Too high gives a flat wide bead and spatter, too low gives a tall narrow bead.

#160 · Level 1 Arc Welding Processes

What does good short circuit welding sound like?

Answer

A steady crackling, like frying bacon. Loud popping or a hissing wire stub means settings are off.

#161 · Level 1 Arc Welding Processes

What is stickout (electrode extension)?

Answer

The length of bare wire from the contact tip to the work.

#162 · Level 1 Arc Welding Processes

What is the usual stickout for short circuit GMAW?

Answer

Commonly 3/8 to 1/2 inch. Keep it steady.

#163 · Level 1 Arc Welding Processes

How does longer stickout affect current at the same wire speed?

Answer

Current drops and penetration falls. Shorter stickout raises current.

#164 · Level 1 Arc Welding Processes

What gun angle is usual for GMAW?

Answer

A push angle of about 5° to 15°, with the nozzle pointing in the direction of travel.

#165 · Level 1 Arc Welding Processes

What effect does pushing have in GMAW?

Answer

A flatter, wider bead, better gas coverage and less penetration than dragging.

#166 · Level 1 Arc Welding Processes

Which vertical progression is generally used on thick plate for strength?

Answer

Uphill, because it gives better penetration and strength. Downhill is shallow.

#167 · Level 1 Arc Welding Processes

How does the FCAW wire differ from a GMAW wire?

Answer

It is a hollow tube filled with flux and alloying powders instead of a solid wire.

#168 · Level 1 Arc Welding Processes

What is FCAW-G?

Answer

Gas-shielded flux cored welding. External gas, usually CO2 or 75/25 argon and CO2, adds to the gas made by the flux, so it is called dual shield.

#169 · Level 1 Arc Welding Processes

What is FCAW-S?

Answer

Self-shielded flux cored welding. The flux alone makes all the shielding, so no gas cylinder is needed.

#170 · Level 1 Arc Welding Processes

What polarity does FCAW-G use?

Answer

DCEP.

#171 · Level 1 Arc Welding Processes

What polarity do many self-shielded FCAW wires use?

Answer

DCEN, but not all do. Read the wire data.

#172 · Level 1 Arc Welding Processes

Why is FCAW-S good for field work in wind?

Answer

It needs no gas cylinder and its flux makes its own shield.

#173 · Level 1 Arc Welding Processes

In E71T-1, what does the T mean?

Answer

Tubular, meaning a flux cored wire.

#174 · Level 1 Arc Welding Processes

In E71T-1, what does the 1 after the 7 mean?

Answer

All positions. A 0 there, as in E70T-1, means flat and horizontal only.

#175 · Level 1 Arc Welding Processes

In E71T-1, what does the final 1 mean?

Answer

The usability designator. It indicates the shielding gas type and performance class, here CO2 shielding.

#176 · Level 1 Arc Welding Processes

What are examples of self-shielded flux cored wires?

Answer

E71T-8 and E71T-11. They are used without shielding gas.

#177 · Level 1 Arc Welding Processes

What does FCAW leave on the weld that GMAW does not?

Answer

Slag, which must be removed between passes like SMAW slag.

#178 · Level 1 Arc Welding Processes

What is the typical stickout for gas-shielded FCAW wire?

Answer

Commonly about 3/4 to 1 inch. Use the manufacturer's range for self-shielded wire.

#179 · Level 1 Arc Welding Processes

Which angle is usual for FCAW?

Answer

A drag angle of 5° to 15°, so the slag trails behind the pool.

#180 · Level 1 Arc Welding Processes

What is the electrode in GTAW?

Answer

A non-consumable tungsten electrode. It does not become part of the weld.

#181 · Level 1 Arc Welding Processes

What shielding gas does GTAW normally use?

Answer

Pure argon, with a typical flow of 15 to 25 CFH.

#182 · Level 1 Arc Welding Processes

When is helium added to GTAW shielding gas?

Answer

When more heat is needed on thick or highly conductive metals.

#183 · Level 1 Arc Welding Processes

What is GTAW best suited for?

Answer

Thin material, stainless steel, aluminum, titanium and critical pipe root passes, where clean precise welds matter.

#184 · Level 1 Arc Welding Processes

Does GTAW produce slag?

Answer

No. It makes clean welds with no slag, but it is slow and needs very clean metal.

#185 · Level 1 Arc Welding Processes

How is current controlled during GTAW?

Answer

With a foot pedal or a thumb switch on the torch.

#186 · Level 1 Arc Welding Processes

What is a high frequency (HF) start?

Answer

The arc jumps to the work without the tungsten touching it, which avoids contamination.

#187 · Level 1 Arc Welding Processes

What is a lift arc start?

Answer

The tungsten lightly touches the work and is lifted to start the arc at low current.

#188 · Level 1 Arc Welding Processes

Why should the tungsten not touch the work to start?

Answer

Touching contaminates the tungsten and makes the arc unstable.

#189 · Level 1 Arc Welding Processes

Which polarity is used for GTAW on steel and stainless steel?

Answer

DCEN. The tungsten stays cooler and can carry high current with a pointed tip.

#190 · Level 1 Arc Welding Processes

Which current is used for GTAW on aluminum and magnesium?

Answer

AC.

#191 · Level 1 Arc Welding Processes

What is cleaning action in AC GTAW?

Answer

During the half cycle with the electrode positive, the arc blasts off the tough oxide layer on aluminum or magnesium.

#192 · Level 1 Arc Welding Processes

What color band marks EWP tungsten?

Answer

Green. EWP is pure tungsten, used on AC for aluminum with a balled tip.

#193 · Level 1 Arc Welding Processes

What is EWTh-2 tungsten?

Answer

2 percent thoriated tungsten with a red band. It starts easily and lasts long on DCEN, but thorium is mildly radioactive.

#194 · Level 1 Arc Welding Processes

What color band marks EWCe-2 tungsten?

Answer

Gray. It is 2 percent ceriated tungsten, good at low current DC and usable on AC.

#195 · Level 1 Arc Welding Processes

What is EWZr-1 tungsten used for?

Answer

Mainly AC welding. It is zirconiated and has a brown band.

#196 · Level 1 Arc Welding Processes

How should a tungsten be ground for DC welding?

Answer

To a point, with the grind marks running lengthwise along the electrode, not around it.

#197 · Level 1 Arc Welding Processes

What is a common taper length for a ground tungsten?

Answer

About two to two and a half times the electrode diameter.

#198 · Level 1 Arc Welding Processes

What do you do if the tungsten touches the weld pool?

Answer

Stop, break off the contaminated end and regrind it. The arc will be unstable until you do.

#199 · Level 1 Arc Welding Processes

How far does tungsten usually extend past the gas cup?

Answer

Commonly about 1/8 to 1/4 inch, and a little more for a hard-to-reach joint.

#200 · Level 1 Arc Welding Processes

What is postflow in GTAW?

Answer

Argon that keeps flowing for several seconds after the arc stops to protect the hot tungsten and cooling weld.

#201 · Level 1 Arc Welding Processes

What rule of thumb is used for postflow time?

Answer

About one second per 10 amps of welding current.

#202 · Level 1 Arc Welding Processes

What does a blue or gray tungsten after welding suggest?

Answer

Postflow was too short or gas coverage was poor. A good tungsten stays bright and silvery.

#203 · Level 1 Arc Welding Processes

What GTAW filler is used for mild steel?

Answer

ER70S-2.

#204 · Level 1 Arc Welding Processes

What GTAW filler is used for 304 stainless steel?

Answer

ER308L.

#205 · Level 1 Arc Welding Processes

What GTAW fillers are used for aluminum?

Answer

ER4043 or ER5356.

#206 · Level 1 Arc Welding Processes

At what angle is GTAW filler rod added?

Answer

A low angle of about 15°, to the leading edge of the pool, keeping the hot rod end inside the gas shield.

#207 · Level 1 Arc Welding Processes

What weld position is 1G?

Answer

Flat position groove weld, made from the top with the pool sitting on top of the joint.

#208 · Level 1 Arc Welding Processes

What weld position is 2F?

Answer

Horizontal fillet, where the weld is on top of a horizontal surface against a vertical one.

#209 · Level 1 Arc Welding Processes

What weld position is 3G?

Answer

Vertical groove weld, running up or down a vertical plate.

#210 · Level 1 Arc Welding Processes

What weld position is 4G?

Answer

Overhead groove weld, made from underneath. It is the most difficult position.

#211 · Level 1 Arc Welding Processes

What do the letters G and F mean in position names?

Answer

G means groove weld and F means fillet weld. The numbers 1 to 4 are the same for both.

#212 · Level 1 Arc Welding Processes

How do you adjust for horizontal position welding?

Answer

Use slightly lower current and a slight upward work angle to fight sag.

#213 · Level 1 Arc Welding Processes

Why is heat input lower in the vertical position?

Answer

Gravity pulls the pool downward, so a smaller, cooler pool is needed.

#214 · Level 1 Arc Welding Processes

How should you weld overhead to keep the pool from dropping?

Answer

Use smaller electrodes or wire, lower current, short arc lengths and stringer or small weave beads.

#215 · Level 1 Arc Welding Processes

What is the heat input at 24 volts, 200 amps and 12 inches per minute?

Answer

24 × 200 × 60 ÷ 12 = 24,000 joules per inch, or 24 kJ/in.

#216 · Level 1 Arc Welding Processes

Why does heat input matter?

Answer

Too much can weaken some steels and cause distortion. Too little leads to cold cracking and lack of fusion.

#217 · Level 1 Arc Welding Processes

What is the difference between a discontinuity and a defect?

Answer

A discontinuity is any break in weld uniformity. A defect is a discontinuity large enough to fail the inspection code.

#218 · Level 1 Arc Welding Processes

How does porosity show up on the surface of a weld?

Answer

Gas pockets trapped in the weld. A surface pore is a pinhole and a line of them is wormhole porosity.

#219 · Level 1 Arc Welding Processes

Which arc welding habits and conditions lead to porosity?

Answer

Damp electrodes, dirty base metal, too long an arc, drafts, empty or clogged gas supply, or too fast travel.

#220 · Level 1 Arc Welding Processes

Which discontinuity is a melted groove left unfilled at the edge of a weld?

Answer

Undercut: a groove melted into the base metal at the weld edge and left unfilled.

#221 · Level 1 Arc Welding Processes

Which welder settings and habits cause undercut?

Answer

Too much current or voltage, too fast travel, too long an arc or the wrong angle.

#222 · Level 1 Arc Welding Processes

What is overlap (cold lap)?

Answer

Weld metal rolls onto the base metal without fusing to it.

#223 · Level 1 Arc Welding Processes

Which discontinuity is weld metal that did not melt into the sidewall or the previous bead?

Answer

Lack of fusion: weld metal that does not melt into the sidewall or the previous bead.

#224 · Level 1 Arc Welding Processes

Which settings and techniques lead to lack of fusion?

Answer

Too little current, too fast travel, poor angle, dirty surfaces or short circuit GMAW at too low a setting.

#225 · Level 1 Arc Welding Processes

Which discontinuity leaves the weld short of the root of the joint?

Answer

Incomplete joint penetration: the weld does not reach the root of the joint.

#226 · Level 1 Arc Welding Processes

What causes incomplete joint penetration in a groove weld, and how is it cured?

Answer

Low current, too large a root face, too small a root gap or fast travel. Cures are more current, wider gap, smaller root face and slower travel.

#227 · Level 1 Arc Welding Processes

Which arc welding processes most often trap slag in the weld?

Answer

Slag trapped in the weld, mostly seen in SMAW and FCAW.

#228 · Level 1 Arc Welding Processes

How do you prevent slag inclusions?

Answer

Chip and brush each pass fully, grind high spots and deep valleys, and keep the arc on the leading edge of the pool.

#229 · Level 1 Arc Welding Processes

What are common causes of weld cracking?

Answer

Hydrogen from wet electrodes, high restraint, rapid cooling, high carbon base metal, too small a bead and unfilled craters.

#230 · Level 1 Arc Welding Processes

How do you prevent cracks?

Answer

Use dry low-hydrogen electrodes, preheat when the WPS requires it, use correct filler, fill craters and avoid excessive restraint.

#231 · Thermal Cutting Processes

What are the three steps of oxy-fuel cutting?

Answer

Preheat flames bring the steel to kindling temperature, then the cutting oxygen stream burns the red-hot iron. The force of the stream blows the iron oxide slag out the bottom of the cut.

#232 · Thermal Cutting Processes

About what temperature must steel reach before the cutting oxygen is used?

Answer

Roughly 1,600°F to 1,800°F, when the steel glows bright cherry red. The preheat flames bring it to that kindling temperature.

#233 · Thermal Cutting Processes

Why can oxy-fuel cutting not cut aluminum in the normal way?

Answer

Aluminum melts at about 1,220°F, but its oxide melts at about 3,700°F. The oxide skin acts like a lid on the cut, so the torch only melts and smears the metal.

#234 · Thermal Cutting Processes

Why is stainless steel not cut with a standard oxy-fuel torch?

Answer

Chromium forms a tough oxide film that resists the burning reaction and melts at a higher temperature than the metal under it. Plasma, saw, laser, or waterjet is used instead.

#235 · Thermal Cutting Processes

What are the holes in a cutting tip for?

Answer

The large center hole carries the cutting oxygen. The ring of smaller holes around it carries the gas mixture for the preheat flames.

#236 · Thermal Cutting Processes

What is slag in oxy-fuel cutting?

Answer

Slag is the iron oxide that the cutting oxygen stream blows out of the bottom of the cut. A small amount that knocks off easily is normal on a good cut.

#237 · Thermal Cutting Processes

What pressure limit applies to acetylene?

Answer

Never set acetylene working pressure above 15 psig. Above that it can break down on its own and explode.

#238 · Thermal Cutting Processes

Why must an acetylene cylinder be used upright?

Answer

Acetylene is dissolved in acetone inside a porous filler. Lying down, the cylinder can let liquid acetone reach the regulator and hose.

#239 · Thermal Cutting Processes

About what pressure does a full oxygen cylinder read?

Answer

About 2,200 psi at room temperature. A reading near 1,100 psi means the cylinder is about half full.

#240 · Thermal Cutting Processes

About what pressure does a full acetylene cylinder read?

Answer

About 250 psi. It is low because acetylene is unstable at higher pressures and is stored dissolved in acetone.

#241 · Thermal Cutting Processes

How far should an acetylene cylinder valve be opened?

Answer

About three quarters of a turn, and never more than one and a half turns. Leave the cylinder wrench on the valve so it can be closed quickly.

#242 · Thermal Cutting Processes

How is an oxygen cylinder valve opened differently from an acetylene valve?

Answer

An oxygen valve is opened all the way until it seats at the back. This seals the valve stem against leaks.

#243 · Thermal Cutting Processes

How can you tell fuel gas fittings and hoses from oxygen ones?

Answer

Fuel gas fittings have left-hand threads with a groove on the nut, and oxygen fittings have right-hand threads. In the US the oxygen hose is green and the fuel gas hose is red.

#244 · Thermal Cutting Processes

What do the two gauges on a regulator show?

Answer

The high pressure gauge shows how much gas is left in the cylinder. The low pressure gauge shows the working pressure going to the hose.

#245 · Thermal Cutting Processes

Why must oil and grease be kept away from oxygen equipment?

Answer

Oil or grease and pure oxygen can ignite on contact. Never touch oxygen cylinders, regulators, or fittings with oily hands or rags.

#246 · Thermal Cutting Processes

How should you test connections for gas leaks?

Answer

Brush on soapy water or leak detection fluid and watch for bubbles. Never use a flame to find a leak.

#247 · Thermal Cutting Processes

How far apart must oxygen and fuel gas cylinders be stored?

Answer

At least 20 feet apart, or separated by a noncombustible barrier at least 5 feet high with a fire rating of at least half an hour.

#248 · Thermal Cutting Processes

What tool should be used to light a cutting torch?

Answer

Use a spark lighter, also called a striker. A match or butane lighter puts your hand too close to the tip.

#249 · Thermal Cutting Processes

In what order are the torch valves closed when shutting off the flame?

Answer

Close the torch fuel valve first, then the oxygen valve. This stops the flame quickly and keeps soot from forming.

#250 · Thermal Cutting Processes

What does a neutral flame look like, and what does it do to the steel?

Answer

It has equal oxygen and acetylene and shows a sharp bright blue inner cone. It neither adds carbon to the steel nor burns it.

#251 · Thermal Cutting Processes

What is a carburizing flame?

Answer

A flame with too much acetylene, shown by a feather between the inner cone and the outer flame. It adds carbon to the steel, making it hard and brittle.

#252 · Thermal Cutting Processes

What is an oxidizing flame?

Answer

A flame with too much oxygen, shown by a short, pointed, hissing inner cone. It can burn the metal and leave a rough surface.

#253 · Thermal Cutting Processes

What is the difference between a backfire and a flashback?

Answer

A backfire is a short pop, and the flame may go out or relight. A flashback is the flame burning back past the mixing point into the torch and hose, with a loud squeal or hiss.

#254 · Thermal Cutting Processes

What should you do first when a flashback occurs?

Answer

Close the torch oxygen valve first, then the fuel valve, then the cylinder valves. Have the equipment inspected before reuse and follow shop procedure.

#255 · Thermal Cutting Processes

What do check valves and flashback arrestors do?

Answer

A check valve lets gas flow only one way so the other gas cannot run backward in the hose. A flashback arrestor stops the flame and often shuts off the gas.

#256 · Thermal Cutting Processes

How does a bevel cut change the thickness the oxygen stream must cut?

Answer

The path length equals the plate thickness divided by the cosine of the bevel angle. A 3/8 inch plate at a 45° bevel is 0.375 ÷ 0.7071 = 0.530 inch.

#257 · Thermal Cutting Processes

What causes drag lines to sweep far backward at the bottom of an oxy-fuel cut?

Answer

Travel speed that is too fast or cutting oxygen pressure that is too low. A good cut has drag lines that are nearly straight up and down.

#258 · Thermal Cutting Processes

What does a plasma arc cutter need that oxy-fuel cutting does not?

Answer

An electrically conductive workpiece and a work clamp on clean bare metal. The transferred arc must complete its circuit through the work.

#259 · Thermal Cutting Processes

How does carbon arc gouging remove metal?

Answer

An arc between a carbon-graphite electrode and the work melts the metal, and compressed air at about 80 to 100 psi blows it out. The electrode is held at about 45° in the travel direction.

#260 · Thermal Cutting Processes

Why is cutting galvanized steel a health risk?

Answer

The zinc coating boils and gives off zinc oxide smoke. Breathing it can cause metal fume fever, a flu-like illness, so use ventilation and a respirator when required.

#261 · Trade Math

How do you add 3/8 and 1/4?

Answer

Give them the same denominator first: 1/4 becomes 2/8. Then 3/8 + 2/8 = 5/8.

#262 · Trade Math

How do you divide by a fraction, for example 5/8 ÷ 1/4?

Answer

Flip the second fraction and multiply. So 5/8 × 4 = 20/8 = 2 1/2.

#263 · Trade Math

What is 7/8 minus 3/16?

Answer

Change 7/8 to 14/16, then subtract: 14/16 minus 3/16 = 11/16.

#264 · Trade Math

What is 2 3/8 + 1 7/8?

Answer

Add wholes (3) and fractions (10/8 = 1 2/8). The total is 4 1/4, or 4.25 in decimals.

#265 · Trade Math

How do you change a fraction to a decimal?

Answer

Divide the top number by the bottom number. For example 5/16 = 5 ÷ 16 = 0.3125.

#266 · Trade Math

What is 3/8 inch as a decimal?

Answer

3 ÷ 8 = 0.375 inch. It is one of the common fractions worth memorizing.

#267 · Trade Math

What is 0.625 as a fraction?

Answer

0.625 = 625/1000, which reduces to 5/8.

#268 · Trade Math

What is 0.1875 as a fraction?

Answer

0.1875 = 1875/10000, which reduces to 3/16.

#269 · Trade Math

What is 7/32 inch as a decimal?

Answer

7 ÷ 32 = 0.21875 inch.

#270 · Trade Math

How many millimeters are in one inch?

Answer

Exactly 25.4 mm. Every inch and metric conversion uses this number.

#271 · Trade Math

How do you convert inches to millimeters?

Answer

Multiply the inches by 25.4. For example 3/8 inch = 0.375 × 25.4 = 9.525 mm.

#272 · Trade Math

How do you convert millimeters to inches?

Answer

Divide the millimeters by 25.4. For example 100 mm ÷ 25.4 = 3.937 inches.

#273 · Trade Math

How do you find a percent of a number?

Answer

Change the percent to a decimal by moving the point two places left, then multiply. So 15% of 80 = 0.15 × 80 = 12.

#274 · Trade Math

How do you find what percent one number is of another?

Answer

Divide the part by the whole, then multiply by 100. If 2.5 lb of a 40 lb spool is wasted, 2.5 ÷ 40 = 0.0625, or 6.25%.

#275 · Trade Math

How do you solve a proportion such as 1/4 = 3/x?

Answer

Cross multiply: multiply the diagonal numbers, then divide by the remaining number. Here x = 3 × 4 ÷ 1 = 12.

#276 · Trade Math

How do you find the perimeter of a rectangle?

Answer

Perimeter = 2 × (length + width). A 24 by 36 inch frame has 2 × 60 = 120 inches, or 10 feet.

#277 · Trade Math

How do you change square inches to square feet?

Answer

Divide by 144, because 12 × 12 = 144. A plate of 864 square inches is 6 square feet.

#278 · Trade Math

What is the formula for the area of a circle?

Answer

Area = π × radius², with π about 3.1416. A circle 6 inches across has area 3.1416 × 9 = 28.27 square inches.

#279 · Trade Math

What is the formula for the volume of a block or plate?

Answer

Volume = length × width × thickness. A plate 12 by 6 by 1/2 inch is 36 cubic inches.

#280 · Trade Math

About how much does carbon steel weigh per cubic inch?

Answer

About 0.283 pound per cubic inch. Multiply volume by this to find the weight of a steel part.

#281 · Trade Math

What is the area of a fillet weld with equal legs?

Answer

The cross section is a right triangle, so area = leg × leg ÷ 2. A 1/4 inch fillet has an area of 0.03125 square inch.

#282 · Trade Math

How much weld metal does a 1/4 inch fillet hold per foot?

Answer

Volume per foot is 0.03125 × 12 = 0.375 cubic inch. At 0.283 lb per cubic inch, that is about 0.106 pound per foot.

#283 · Trade Math

How much weld metal does a 3/8 inch fillet hold per foot?

Answer

Area is 0.0703 square inch, volume per foot is 0.844 cubic inch, and weight is about 0.239 pound per foot.

#284 · Trade Math

How much weld metal does a 1/2 inch fillet hold per foot?

Answer

Area is 0.125 square inch, volume per foot is 1.5 cubic inches, and weight is about 0.425 pound per foot.

#285 · Trade Math

What happens to weld metal if you double the fillet leg size?

Answer

The weld metal quadruples, because area depends on the leg multiplied by itself. A 1/2 inch fillet holds 4 times as much as a 1/4 inch fillet.

#286 · Trade Math

How much weld metal is in 50 feet of 1/4 inch fillet?

Answer

About 0.106 × 50 = 5.3 pounds of weld metal.

#287 · Trade Math

What is deposition efficiency?

Answer

It is the percent of the filler metal used that actually ends up in the weld. The rest is lost as spatter, slag, stubs or fume.

#288 · Trade Math

How do you find how much filler to buy for a given weld metal weight?

Answer

Divide the weld metal weight by the efficiency as a decimal. For 5.3 lb at 95% efficiency, 5.3 ÷ 0.95 = 5.58 lb of wire.

#289 · Trade Math

How much stick electrode is needed for 5.3 pounds of weld metal at 60% efficiency?

Answer

5.3 ÷ 0.60 = 8.83 pounds of electrodes. Stick efficiency is low because of the stub and flux coating.

#290 · Trade Math

How is a bevel angle measured when laying out a cut on plate?

Answer

It is the angle of the cut edge, measured from a square cut across the plate edge. A 30 degree bevel leaves a face 60 degrees from the plate surface.

#291 · Trade Math

How are bevel angle and included angle related when both plates are beveled equally?

Answer

The included angle is twice the bevel angle. A 30 degree bevel on each plate gives a 60 degree included angle.

#292 · Trade Math

What cut angle does each piece of a two piece 90 degree miter need?

Answer

Each cut is half of the turn, so 90 ÷ 2 = 45 degrees.

#293 · Trade Math

What is the cut angle for each cut in a three piece 90 degree elbow?

Answer

The 90 degree turn is split over two joints, so 45 degrees per joint. Each joint is two cuts, so each cut is 22.5 degrees.

#294 · Trade Math

What is the formula for the circumference of a pipe?

Answer

C = π × D, using the true outside diameter. NPS 6 pipe has an outside diameter of 6.625 inches, so C = 20.81 inches.

#295 · Trade Math

What is the outside diameter of NPS 6 pipe?

Answer

6.625 inches. For common pipe sizes the true outside diameter is larger than the nominal size, so use the real OD in formulas.

#296 · Trade Math

How do you find a pipe diameter from a measured circumference?

Answer

Divide the circumference by π. A circumference of 31.4 inches gives 31.4 ÷ 3.1416 = 10.0 inches.

#297 · Trade Math

How do you convert wire feed speed to pounds per hour?

Answer

Find the weight per inch of wire (π ÷ 4 × diameter² × 0.283), multiply by inches per minute, then by 60. At 400 ipm, 0.035 inch wire melts about 6.53 lb per hour.

#298 · Trade Math

How much 0.045 inch wire is melted per hour at 300 ipm?

Answer

Weight per inch is 0.000450 lb. Then 300 × 0.000450 × 60 = 8.10 pounds per hour, or about 7.70 deposited at 95% efficiency.

#299 · Trade Math

How do you calculate travel speed?

Answer

Divide the weld length by the time. Welding 12 inches in 48 seconds (0.8 minute) gives 12 ÷ 0.8 = 15 inches per minute.

#300 · Trade Math

What is the heat input formula in kJ per inch?

Answer

Heat input = (volts × amps × 60) ÷ (travel speed in inches per minute × 1000). At 24 V, 220 A and 12 in/min, heat input is 26.4 kJ/in. Doubling travel speed cuts it in half.

#301 · Welding Metallurgy

What is the carbon range of low carbon (mild) steel?

Answer

Low carbon steel has up to about 0.30% carbon. It welds easily and includes most structural steel such as A36 plate.

#302 · Welding Metallurgy

What is the carbon range of medium carbon steel?

Answer

Medium carbon steel has about 0.30% to 0.60% carbon. It is stronger but usually needs preheat to avoid cracking.

#303 · Welding Metallurgy

Why is high carbon steel hard to weld?

Answer

High carbon steel has about 0.60% to 1.0% carbon or more. It turns hard and brittle after welding, so it needs careful preheat and slow cooling.

#304 · Welding Metallurgy

What does carbon do to steel?

Answer

Carbon makes steel stronger and harder. It also lowers ductility and makes the steel harder to weld.

#305 · Welding Metallurgy

What makes a steel a stainless steel?

Answer

Stainless steel contains at least about 10.5% chromium. The chromium forms a thin protective oxide layer on the surface.

#306 · Welding Metallurgy

Why are chromium and molybdenum added to low alloy steels?

Answer

They add strength at high temperature. That is why they are common in boilers and pressure piping.

#307 · Welding Metallurgy

What does hardenability mean?

Answer

Hardenability is how easily a steel turns hard when it is cooled quickly. Manganese and other alloying elements raise it.

#308 · Welding Metallurgy

What is carbon equivalent (CE)?

Answer

CE adds the effect of carbon and other hardening elements into one number as if they were all carbon. A higher CE means more risk of a hard, crack-prone weld zone.

#309 · Welding Metallurgy

What is the IIW carbon equivalent formula?

Answer

CE = C + Mn ÷ 6 + (Cr + Mo + V) ÷ 5 + (Ni + Cu) ÷ 15. Each symbol is the percent of that element in the steel.

#310 · Welding Metallurgy

Find CE for C 0.20, Mn 0.90, Cr 0.10, Mo 0.05, V 0.02, Ni 0.10, Cu 0.15.

Answer

Mn ÷ 6 is 0.15, (Cr + Mo + V) ÷ 5 is 0.034 and (Ni + Cu) ÷ 15 is 0.0167. Added to 0.20 carbon, CE is about 0.40.

#311 · Welding Metallurgy

What does a low CE tell you about weldability?

Answer

A low CE, roughly below 0.35 to 0.40, means the steel is easy to weld and often needs little preheat on thin sections. Always check the code and the WPS for the real limit.

#312 · Welding Metallurgy

What is the heat-affected zone (HAZ)?

Answer

The HAZ is base metal that did not melt but got hot enough for its internal structure to change. It sits between the fusion line and the unaffected base metal.

#313 · Welding Metallurgy

Which part of the HAZ is usually hardest and most brittle?

Answer

The coarse grain region next to the fusion line. It got the hottest and its grains grew large.

#314 · Welding Metallurgy

Write the arc welding heat input formula.

Answer

Heat input in kJ per inch = (volts × amps × 60) ÷ (travel speed in inches per minute × 1000).

#315 · Welding Metallurgy

Find heat input at 25 V, 200 A and 10 inches per minute.

Answer

25 × 200 × 60 = 300,000. Divided by 10 × 1000 gives 30 kJ per inch.

#316 · Welding Metallurgy

What happens to heat input and HAZ width when travel speed increases?

Answer

Heat input drops and the HAZ gets narrower. Cooling is also faster.

#317 · Welding Metallurgy

What are grains in metal?

Answer

Grains are small crystals in which the atoms sit in an orderly pattern. Grain boundaries are the borders between them.

#318 · Welding Metallurgy

How does grain size affect toughness?

Answer

Small, fine grains give a tougher metal. Large, coarse grains give a more brittle one.

#319 · Welding Metallurgy

Why does thick plate cool faster than thin plate?

Answer

Heat flows away from the weld in three directions in thick plate. That faster cooling makes hardening and cracking more likely.

#320 · Welding Metallurgy

Name three things that speed up cooling after welding.

Answer

Thick plate, cold plate and low heat input all speed up cooling. Small stringer beads on thick metal also cool quickly.

#321 · Welding Metallurgy

What is the main job of preheat?

Answer

Preheat slows the cooling rate. That gives softer structures, lowers cracking risk and lets hydrogen escape.

#322 · Welding Metallurgy

Name conditions that call for more preheat.

Answer

Higher carbon or CE, thicker plate, rigid joints and a cold shop all call for more preheat. High hydrogen conditions do too.

#323 · Welding Metallurgy

What is interpass temperature?

Answer

It is the temperature of the weld area between passes. Procedures usually set a minimum and a maximum.

#324 · Welding Metallurgy

Why is a maximum interpass temperature set?

Answer

Too much heat builds up, grains grow and strength and toughness drop. The maximum protects the properties of the joint.

#325 · Welding Metallurgy

Where should preheat temperature be measured?

Answer

Measure on the base metal close to the joint, preferably on the side away from the heat source. That shows the whole thickness has soaked.

#326 · Welding Metallurgy

What tools check preheat and interpass temperature?

Answer

Temperature indicating crayons, contact pyrometers and infrared thermometers. A crayon mark melts at its stated temperature.

#327 · Welding Metallurgy

What is ferrite?

Answer

Ferrite is body centered cubic iron that holds at most about 0.02% carbon. It is soft, ductile and magnetic.

#328 · Welding Metallurgy

What is austenite?

Answer

Austenite is face centered cubic iron that forms when steel is heated above its critical temperature. It can dissolve far more carbon and is not magnetic.

#329 · Welding Metallurgy

At about what temperature does 0.77% carbon steel start changing to austenite?

Answer

About 1341°F (727°C). Steels with less carbon finish changing at higher temperatures.

#330 · Welding Metallurgy

What is cementite?

Answer

Cementite is iron carbide, a very hard, brittle compound with 6.67% carbon.

#331 · Welding Metallurgy

What is pearlite?

Answer

Pearlite is a layered mixture of ferrite and cementite. It is strong and moderately ductile.

#332 · Welding Metallurgy

What is martensite and why is it a welding problem?

Answer

Martensite forms when austenite cools too fast and traps carbon in a stressed, distorted pattern. It is very hard and brittle, so it cracks easily.

#333 · Welding Metallurgy

How does cooling speed decide the structure in steel?

Answer

Slow cooling gives soft pearlite. Medium cooling gives tough bainite and fast cooling gives hard martensite.

#334 · Welding Metallurgy

Define tempering.

Answer

Tempering is reheating hardened steel to a moderate temperature. It softens the steel a little and lifts toughness.

#335 · Welding Metallurgy

What is the difference between tensile strength and yield strength?

Answer

Tensile strength is the highest stress the metal can carry before breaking. Yield strength is the stress at which it starts to bend permanently.

#336 · Welding Metallurgy

What is ductility and how is it often reported?

Answer

Ductility is how far metal stretches before breaking. It is often given as percent elongation.

#337 · Welding Metallurgy

What does a Charpy V-notch test measure?

Answer

It measures toughness, the energy a notched metal bar absorbs when struck. Toughness drops as temperature falls.

#338 · Welding Metallurgy

Find stress in a 0.5 inch by 1 inch bar pulled with 18,000 pounds.

Answer

Area is 0.5 in². Stress is 18,000 ÷ 0.5 = 36,000 psi, or 36 ksi.

#339 · Welding Metallurgy

What is the minimum yield strength of A36 steel?

Answer

A36 steel has a minimum yield of 36 ksi.

#340 · Welding Metallurgy

What is postweld heat treatment (PWHT)?

Answer

PWHT is controlled heating of a finished weld. The part is heated slowly, held at temperature and cooled slowly.

#341 · Welding Metallurgy

What are typical stress relief conditions for carbon steel?

Answer

About 1100°F to 1250°F, held about one hour per inch of thickness. The code sets the real values.

#342 · Welding Metallurgy

Name benefits of stress relief.

Answer

It lowers residual stress, softens the hard HAZ, improves toughness and helps hydrogen escape.

#343 · Welding Metallurgy

What is residual stress?

Answer

Residual stress is stress left inside a metal with no outside load. In a restrained weld it can reach the yield strength.

#344 · Welding Metallurgy

Why does a weld distort?

Answer

The hot weld metal wants to expand but is held back by cold metal. On cooling it shrinks and pulls the plate with it.

#345 · Welding Metallurgy

Give three ways to control distortion.

Answer

Use the smallest weld size that meets the drawing, balance welds on both sides and use a backstep or skip sequence. Clamps, jigs and presetting also help.

#346 · Welding Metallurgy

What is the trade off of heavy clamping?

Answer

Clamping reduces distortion but locks in more residual stress. On hard steels cracking risk can go up.

#347 · Welding Metallurgy

What three conditions are needed for hydrogen cracking?

Answer

Hydrogen in the weld, a hard microstructure such as martensite and tensile stress. Removing any one stops the cracking.

#348 · Welding Metallurgy

When does hydrogen cracking appear?

Answer

It is delayed, often hours or a day or more after welding. It tends to occur below about 300°F.

#349 · Welding Metallurgy

What do the digits in E7018 mean?

Answer

The 70 is minimum tensile strength of 70 ksi, the 1 is all positions and the 8 is a low hydrogen iron powder covering. Check the AWS classification for current types.

#350 · Welding Metallurgy

How should low hydrogen electrodes be stored once the can is open?

Answer

Keep them in a heated rod oven, often about 250°F to 300°F. Follow the WPS and maker for exposure limits.

#351 · Welding Metallurgy

Why is E6010 a high hydrogen electrode?

Answer

Its cellulose covering puts a lot of hydrogen into the weld. It is not a low hydrogen electrode.

#352 · Welding Metallurgy

What is sensitization in stainless steel?

Answer

In austenitic stainless held at roughly 800°F to 1500°F, chromium carbides form at the grain boundaries. That uses up nearby chromium and allows intergranular corrosion.

#353 · Welding Metallurgy

Name two ways to prevent sensitization.

Answer

Use low carbon grades such as 304L or stabilized grades such as 321 and 347. Low heat input and controlled interpass temperature also help.

#354 · Welding Metallurgy

Which fillers are used for 304, 316 and carbon steel to stainless?

Answer

ER308L is typical for 304 and ER316L for 316. ER309L is used for carbon steel to stainless.

#355 · Welding Metallurgy

Why does austenitic stainless distort more than carbon steel?

Answer

It expands about 1.5 times as much and conducts only about one third as much heat. Heat stays near the weld.

#356 · Welding Metallurgy

Why is aluminum oxide a welding problem?

Answer

The oxide melts at about 3700°F while the aluminum melts at about 1220°F. It must be removed just before welding, and AC on GTAW helps break it up.

#357 · Welding Metallurgy

Which common aluminum fillers are used and for what?

Answer

ER4043 has silicon and resists cracking in 6061. ER5356 has magnesium and gives more strength on 5xxx alloys.

#358 · Welding Metallurgy

What is the carbon range and main weld problem of cast iron?

Answer

Cast iron has about 2% to 4% carbon. It is brittle and cannot stretch to absorb shrinkage, so it cracks.

#359 · Welding Metallurgy

How is gray cast iron usually welded?

Answer

Preheat, short beads with light peening, keep it warm between beads and cool slowly. Nickel base electrodes such as ENi-CI are common.

#360 · Welding Metallurgy

Why use ER309L for dissimilar welds and what is dilution?

Answer

Dilution is base metal mixing into the weld pool. ER309L has extra chromium and nickel to handle dilution from carbon steel and avoid a brittle weld.

#361 · Welding Inspection and Testing

What is VT in welding inspection?

Answer

VT stands for visual testing, or visual inspection. It is the most used inspection method and costs almost nothing. Most codes require it before any other test.

#362 · Welding Inspection and Testing

Name the three stages of visual inspection.

Answer

Visual inspection is done before welding, during welding and after welding. Each stage checks different things, from fit-up to pass cleaning to final weld size and shape.

#363 · Welding Inspection and Testing

What is checked in visual inspection before welding?

Answer

The inspector checks base metal type, joint angle, root opening, fit-up, alignment and cleanliness. Filler metal and any required preheat are checked too.

#364 · Welding Inspection and Testing

What is checked during welding?

Answer

The inspector checks pass size, slag removal between passes, interpass temperature and pass sequence. Problems caught here are easy to fix before the next pass buries them.

#365 · Welding Inspection and Testing

What is checked after welding?

Answer

The inspector checks weld size, shape, length and surface discontinuities. Everything is compared with the acceptance rules in the code or contract.

#366 · Welding Inspection and Testing

What does a fillet weld gauge measure?

Answer

It measures the leg size of a fillet weld using steel leaves notched for set sizes. Some gauges also measure convexity and concavity.

#367 · Welding Inspection and Testing

What does a bridge cam gauge read?

Answer

It reads reinforcement height, root opening, bevel angle and undercut depth on groove welds. It is also called a Cambridge gauge.

#368 · Welding Inspection and Testing

What does a Hi-Lo gauge check?

Answer

It checks misalignment between two pieces, especially pipe ends. It shows whether the inside edges line up.

#369 · Welding Inspection and Testing

How is the throat of an ideal equal-leg fillet found from its leg?

Answer

Throat equals about 0.707 times the leg length. For a 3/8 inch leg the throat is 0.375 × 0.707, which is about 0.265 inch.

#370 · Welding Inspection and Testing

Why is throat more important than leg in a fillet weld?

Answer

Strength depends on the throat, the shortest distance from root to face. A bulging fillet can look large but still have a thin throat.

#371 · Welding Inspection and Testing

Define discontinuity.

Answer

A discontinuity is any interruption in the normal structure of a weld, such as a pore or slag bit. Having one does not by itself make a weld bad.

#372 · Welding Inspection and Testing

Define defect.

Answer

A defect is a discontinuity large enough, or of a harmful enough type, to fail the acceptance standard. A defect must be repaired or the weld rejected.

#373 · Welding Inspection and Testing

Where should acceptance limits come from?

Answer

They come from the governing code or contract, never from memory. Limits differ for a bridge, a pressure vessel and a handrail.

#374 · Welding Inspection and Testing

What is porosity?

Answer

Porosity is gas trapped in the weld metal as tiny holes. It may be scattered, clustered or lined up.

#375 · Welding Inspection and Testing

What causes porosity?

Answer

Common causes are dirty or damp base metal, damp electrodes, too little shielding gas, drafts, a clogged nozzle and a long arc.

#376 · Welding Inspection and Testing

How is porosity cured?

Answer

Clean the joint, keep electrodes dry, check gas flow and connections, block drafts and shorten the arc.

#377 · Welding Inspection and Testing

What is a slag inclusion?

Answer

It is slag trapped inside the weld. It occurs in stick, flux cored and submerged arc welding.

#378 · Welding Inspection and Testing

How are slag inclusions prevented?

Answer

Chip and brush every pass thoroughly, use proper current, and keep the bead flat enough that slag can float out.

#379 · Welding Inspection and Testing

What is lack of fusion?

Answer

Lack of fusion is weld metal that did not melt into the base metal or the previous bead. It can occur at a side wall, between passes or at the root.

#380 · Welding Inspection and Testing

What causes lack of fusion?

Answer

Too little heat, travel that is too fast, a wrong electrode angle, a dirty surface or weaving so wide that metal skips the side wall.

#381 · Welding Inspection and Testing

What is incomplete joint penetration?

Answer

The weld metal did not reach the full depth the joint required, often leaving the root unwelded.

#382 · Welding Inspection and Testing

What causes incomplete joint penetration?

Answer

A root opening that is too small, a root face that is too thick, low current, or an electrode too large to reach the root.

#383 · Welding Inspection and Testing

What is undercut?

Answer

Undercut is a groove melted into the base metal along the toe of the weld and left unfilled.

#384 · Welding Inspection and Testing

What causes undercut?

Answer

Current that is too high, a long arc, travel that is too fast and a poor electrode angle. Pausing at the sides when weaving helps cure it.

#385 · Welding Inspection and Testing

What is overlap?

Answer

Overlap, or cold lap, is weld metal that rolled over the base metal surface without fusing to it.

#386 · Welding Inspection and Testing

What causes overlap?

Answer

Travel that is too slow and current that is too low. Metal piles up and falls over the edge.

#387 · Welding Inspection and Testing

Why are cracks the most serious discontinuity?

Answer

Codes generally allow no cracks at all, and cracks can spread under load. A crack must be ground out completely, confirmed gone by testing, then rewelded.

#388 · Welding Inspection and Testing

Name common causes of weld cracking.

Answer

Hydrogen in the weld, high restraint, rapid cooling, high carbon or sulfur in the base metal, an undersized weld and an unfilled crater.

#389 · Welding Inspection and Testing

What is a crater crack and how is it avoided?

Answer

A crater crack starts in the crater at the end of a bead. Fill the crater before breaking the arc.

#390 · Welding Inspection and Testing

What is an arc strike?

Answer

An arc strike is a spot on the base metal where an arc was struck outside the weld joint. It can leave a hard, crack-prone spot, so many codes reject it.

#391 · Welding Inspection and Testing

What is burn-through?

Answer

Burn-through is a hole melted through the base metal. It is common on thin material when heat is too high.

#392 · Welding Inspection and Testing

Why is excess reinforcement not extra strength?

Answer

A weld face built up too high creates a stress concentration at the toes. It adds stress, not strength.

#393 · Welding Inspection and Testing

What is the main limit of liquid penetrant testing?

Answer

PT finds only flaws open to the surface. A subsurface flaw is not detected at all.

#394 · Welding Inspection and Testing

List the steps of liquid penetrant testing.

Answer

Clean and dry, apply penetrant, wait the dwell time, remove excess, apply developer, inspect, then clean the part.

#395 · Welding Inspection and Testing

What is dwell time in PT?

Answer

Dwell time is the period the penetrant needs to soak into flaws. It comes from the procedure and varies with material and temperature.

#396 · Welding Inspection and Testing

What is the difference between visible and fluorescent penetrant?

Answer

Visible penetrant is usually red and is seen in normal light. Fluorescent penetrant glows under ultraviolet light in a darkened area and is more sensitive.

#397 · Welding Inspection and Testing

Which materials can PT test that MT cannot?

Answer

PT works on any nonporous material, including stainless steel and aluminum. MT works on ferromagnetic metals only.

#398 · Welding Inspection and Testing

Which metals can be tested by magnetic particle testing?

Answer

Only ferromagnetic metals, those a magnet attracts, such as carbon steel and low alloy steel. Austenitic stainless and aluminum cannot be tested.

#399 · Welding Inspection and Testing

How does MT reveal a flaw?

Answer

A flaw interrupts the magnetic field and makes it leak out at the surface. Iron particles gather at the leak and outline the flaw.

#400 · Welding Inspection and Testing

What flaw direction does MT find best?

Answer

MT finds flaws best when they lie at a right angle to the magnetic field. A flaw parallel to the field barely disturbs it.

#401 · Welding Inspection and Testing

Why is a part magnetized in two directions for MT?

Answer

Flaws can run any way, and each field direction finds flaws across it. Using both directions covers more flaw orientations.

#402 · Welding Inspection and Testing

What field does current passed through a part create?

Answer

Passing current through the part or between prods creates a circular field around the current path. A coil creates a longitudinal field along the length.

#403 · Welding Inspection and Testing

What is a yoke in MT?

Answer

A yoke is a handheld electromagnet that puts a field between its two legs. It is the common tool for weld inspection.

#404 · Welding Inspection and Testing

What does ultrasonic testing use to find flaws?

Answer

UT uses very high frequency sound pulses sent by a transducer. Reflections from flaws or the back wall return and show distance on a screen.

#405 · Welding Inspection and Testing

Why is couplant used in UT?

Answer

Couplant is a gel between the probe and the metal. It lets sound pass because air blocks it.

#406 · Welding Inspection and Testing

Which probe is normally used on welds in UT?

Answer

An angle beam probe sending a shear wave at an angle, usually 45, 60 or 70 degrees. It reaches across the weld from the side.

#407 · Welding Inspection and Testing

How is a UT instrument calibrated?

Answer

It is calibrated on a reference block before use. The block lets the operator set the screen and check response.

#408 · Welding Inspection and Testing

How does radiographic testing make a picture?

Answer

A source sends radiation through the weld to film or a digital detector on the far side. Thicker areas look lighter and thinner areas look darker.

#409 · Welding Inspection and Testing

What do common flaws look like on a radiograph?

Answer

A pore is a dark round spot, slag is a dark irregular spot, and a crack is a thin dark line. Lack of fusion is usually a dark line along the weld edge.

#410 · Welding Inspection and Testing

What sources are used in field radiography?

Answer

An X-ray machine or a radioactive isotope, most often iridium-192 or cobalt-60. Only certified personnel perform RT.

#411 · Welding Inspection and Testing

What is an image quality indicator?

Answer

Also called a penetrameter, it is placed on the part during RT. If its required wire or hole shows on the film, the technique is sensitive enough.

#412 · Welding Inspection and Testing

What three controls keep radiography safe?

Answer

Limit time near the source, stay as far away as possible, and use shielding.

#413 · Welding Inspection and Testing

What is the difference between surface and volumetric NDT methods?

Answer

PT and MT are surface methods. UT and RT are volumetric because they look through the whole volume of the weld.

#414 · Welding Inspection and Testing

Why is a guided bend test used?

Answer

It checks soundness and ductility of a welder's test weld. The coupon is bent around a die into a U shape and the convex surface is examined.

#415 · Welding Inspection and Testing

Compare face, root and side bends.

Answer

A face bend puts the weld face on the outside of the curve, and a root bend puts the root there. A side bend puts the cut cross section outside and is used for thicker plate.

#416 · Welding Inspection and Testing

How is tensile strength calculated?

Answer

Divide the maximum load by the original cross section area. For example, 27,000 lb on 0.375 in^2 gives 72,000 psi.

#417 · Welding Inspection and Testing

What does a fillet weld break test show?

Answer

The fillet is forced to fail at its root, and the fracture is examined for fusion, penetration, porosity and slag.

#418 · Welding Inspection and Testing

What does a macroetch test show?

Answer

A cross section ground smooth and etched with acid shows passes, sidewall fusion, root penetration, heat affected zone, leg length and throat.

#419 · Welding Inspection and Testing

What does the Charpy impact test measure?

Answer

It measures toughness, the energy a notched bar absorbs when hit by a pendulum, in foot pounds. It is done at a stated temperature and used for procedure qualification.

#420 · Welding Inspection and Testing

Why are test coupon corners rounded and reinforcement removed before a bend test?

Answer

Reinforcement is ground flush and corners are rounded slightly so the edge does not start a crack. Grind marks across the specimen can look like cracks.

#421 · Governing Documents and Codes

What is a code in welding?

Answer

A code is a set of rules written to protect safety and quality for a certain kind of work. It becomes a legal or contract requirement when a government, owner, or contract says it applies. It says what must be done and what results are acceptable.

#422 · Governing Documents and Codes

What is a standard?

Answer

A standard is an agreed document that sets out requirements, test methods, sizes, or practices. Groups such as AWS, ASME, and API write many of them. A code is a standard that has been adopted as the rule for a certain job.

#423 · Governing Documents and Codes

What is a specification in welding work?

Answer

A specification describes the requirements for a particular material, product, or job. A filler metal specification, for example, sets the makeup and strength an electrode must have. It is about the thing being bought or made.

#424 · Governing Documents and Codes

What is a procedure?

Answer

A procedure is a step by step instruction for how to do something. The welding procedure specification turns the broad rules of a code into exact instructions for one weld. It tells the welder how to do the work.

#425 · Governing Documents and Codes

A short way to separate code, specification, and procedure?

Answer

The code says what must be true. The specification says what the material or job must be. The procedure says how to do the work.

#426 · Governing Documents and Codes

What does AWS D1.1 cover?

Answer

AWS D1.1 is the Structural Welding Code for steel. It covers welded steel structures such as building frames. It includes prequalified procedures, qualification of procedures and welders, fabrication, and inspection.

#427 · Governing Documents and Codes

What does AWS D1.5 cover?

Answer

AWS D1.5 is the Bridge Welding Code, used for welded highway bridges. Bridges carry moving loads that repeat many times, so the code pays close attention to fatigue and fracture critical members. It is stricter than a building code in several areas.

#428 · Governing Documents and Codes

What is a fracture critical member?

Answer

It is a bridge part whose failure could bring down the bridge or a large part of it. Codes such as AWS D1.5 set extra rules for welding these parts. The extra rules exist because a hidden flaw could grow under repeated loading.

#429 · Governing Documents and Codes

What does API 1104 cover?

Answer

API 1104 is the standard for welding pipelines and related facilities. It covers qualification of procedures and welders, production welding, and the standards for judging flaws found by inspection. Many pipeline welds are made with the pipe fixed in place.

#430 · Governing Documents and Codes

What does ASME B31.3 cover?

Answer

ASME B31.3 is the Process Piping code, used for piping in refineries, chemical plants, and similar facilities. It sets design, material, fabrication, examination, and testing rules. It points to ASME Section IX for welding qualification.

#431 · Governing Documents and Codes

What does ASME Section IX cover?

Answer

Section IX, Welding, Brazing, and Fusing Qualifications, is part of the ASME Boiler and Pressure Vessel Code. It tells how to qualify welding procedures and welders. It does not tell you how to build the vessel or piping.

#432 · Governing Documents and Codes

What is a WPS?

Answer

A welding procedure specification is the written instruction a welder follows. It lists items such as the process, base metal, joint, filler metal, shielding gas, preheat, position, and electrical settings. The welder works to it the way a cook works to a recipe.

#433 · Governing Documents and Codes

What is a PQR?

Answer

A procedure qualification record is the record of a test weld and its test results. It lists the actual values used and the results of tests such as bends and tensile tests. It proves that those settings can make a sound weld.

#434 · Governing Documents and Codes

What is a WPQ?

Answer

A welder performance qualification is the record that proves a particular welder can make a sound weld. The welder makes a test weld that is judged by visual inspection and bend tests or radiography. A pass qualifies the welder for the ranges listed on the record.

#435 · Governing Documents and Codes

What is the difference between a PQR and a WPS?

Answer

A PQR records what was actually done and tested on a test piece. A WPS is the instruction for production work, with working ranges based on the PQR. The PQR proves the procedure and the WPS applies it.

#436 · Governing Documents and Codes

What does each document in the chain prove or do?

Answer

The PQR proves the procedure works. The WPS tells people how to apply it. The WPQ proves the welder can follow it.

#437 · Governing Documents and Codes

Does every WPS need its own PQR test?

Answer

Under ASME Section IX, a WPS must be supported by a PQR. Under AWS D1.1, a WPS that stays within the strict limits of the prequalified section needs no PQR test. A WPS outside those limits must be qualified by testing.

#438 · Governing Documents and Codes

What is an essential variable?

Answer

An essential variable is a change that could affect the mechanical properties of the weld, such as strength or toughness. If it changes beyond the code limits, the procedure must be qualified again with a new test and PQR.

#439 · Governing Documents and Codes

Give examples of changes that are often essential variables.

Answer

Examples include changing the welding process, the base metal group, the filler metal classification, a large change in preheat, or going outside the tested thickness range. Check the code in use for the exact limits.

#440 · Governing Documents and Codes

What is a nonessential variable?

Answer

It is a change not expected to affect the mechanical properties of the weld. The WPS can be revised for such a change without a new test. The WPS document must still be updated to match.

#441 · Governing Documents and Codes

What is a supplementary essential variable?

Answer

It matters only when the code requires toughness testing, such as for low temperature service. When toughness is required, it acts like an essential variable. When toughness is not required, it is treated as nonessential.

#442 · Governing Documents and Codes

What happens to a welder's qualification if they stop welding with a process for a long time?

Answer

The qualification can lapse. The code states how long a gap is allowed, and the period differs between codes. Check the code in use for the exact time.

#443 · Governing Documents and Codes

What do the letters G and F mean in a position name like 3G or 2F?

Answer

G means a groove weld and F means a fillet weld. The number shows the position: 1 flat, 2 horizontal, 3 vertical, and 4 overhead.

#444 · Governing Documents and Codes

What are the four plate groove positions?

Answer

They are 1G flat, 2G horizontal, 3G vertical, and 4G overhead. Gravity affects the molten metal differently in each. The harder positions are overhead and vertical.

#445 · Governing Documents and Codes

What is the 2G position on pipe?

Answer

The pipe stands upright with its axis vertical, and the weld runs around it horizontally. The pipe is not turned during welding.

#446 · Governing Documents and Codes

What is the 5G position on pipe?

Answer

The pipe is fixed horizontal and cannot be turned. The welder works around it through the flat, vertical, and overhead areas.

#447 · Governing Documents and Codes

What is the 6G position on pipe?

Answer

The pipe is fixed at an angle of 45° to the horizontal and is not turned. It is a hard test because the welder must weld in all areas on one fixed joint.

#448 · Governing Documents and Codes

Does qualifying in a harder position cover easier ones?

Answer

Often it does, but the exact rule depends on the code and the joint type. Generally a pass in the vertical and overhead plate positions covers all plate positions. Always read the code tables rather than assuming.

#449 · Governing Documents and Codes

What is the order of precedence in contract documents?

Answer

It is the list in the contract that says which document wins when two documents disagree. Arrangements differ between contracts. Read the clause in your own contract and never guess.

#450 · Governing Documents and Codes

What should be done if two contract documents conflict?

Answer

Ask the responsible person, often through a written request for information, and wait for a written answer. Do not guess which document to follow. Guessing is how costly mistakes happen.

#451 · Governing Documents and Codes

Which documents are usually part of the contract set for a welding job?

Answer

They usually include the contract itself, the drawings, the project specifications, the referenced codes and standards, and changes such as addenda and change orders.

#452 · Governing Documents and Codes

What are the main duties of a welder under the rules?

Answer

The welder follows the WPS that applies, uses the correct materials and equipment, and welds only within the limits of the qualification held. The welder also cleans between passes, reports problems, and follows safety rules.

#453 · Governing Documents and Codes

What are the main duties of a welding inspector?

Answer

The inspector checks the documents, materials, welder qualifications, and the procedure in use. The inspector examines the work before, during, and after welding and reports whether it meets the code. The inspector does not weld the joint.

#454 · Governing Documents and Codes

Who carries the overall duty to make welds meet the contract?

Answer

The contractor or fabricator who employs the welder carries it. This includes qualifying procedures and welders, controlling the work, and repairing rejected welds. The owner may verify that the work meets the requirements.

#455 · Governing Documents and Codes

What does SENSE stand for?

Answer

SENSE stands for Schools Excelling through National Skills Education. It is an American Welding Society program that sets a national standard for welding education and training. This book prepares students for Level 1 and Level 2.

#456 · Governing Documents and Codes

What are QC10 and QC11 in the SENSE program?

Answer

They are numbers of AWS documents that support the entry level and advanced level qualification and certification work in the program. Know them as labels. Your instructor will say exactly which one applies to which level.

#457 · Governing Documents and Codes

What does the SENSE Assessment Coordinator do?

Answer

The Assessment Coordinator is a person at the school recognized by the program to manage testing. The coordinator makes sure exams and performance tests are given under the proper rules, records are kept, and results are reported correctly.

#458 · Governing Documents and Codes

What score is needed to pass a SENSE module knowledge exam?

Answer

A student needs at least 75 percent. On a 40 question exam, that is 0.75 × 40 = 30 correct answers. A lower score means the module exam is not passed.

#459 · Governing Documents and Codes

What score is needed on the SENSE safety exam, and why?

Answer

The safety exam requires 100 percent. A single safety mistake can injure or kill someone, so the program does not accept a safety score with wrong answers. A student who misses any question must study again and retake it.

#460 · Governing Documents and Codes

What is a prequalified WPS?

Answer

It is a WPS that meets the strict limits set out in the prequalified section of AWS D1.1, so no procedure test is needed. If any part falls outside those limits, the procedure must be qualified by testing and a PQR.

#461 · Level 2 Plate and Pipe Welding

What does the G in 1G, 2G, 3G and 4G stand for?

Answer

G stands for groove weld. F would stand for fillet. The number is the welding position.

#462 · Level 2 Plate and Pipe Welding

Describe the 2G plate position.

Answer

The plate stands upright and the weld runs sideways across it. This is the horizontal position.

#463 · Level 2 Plate and Pipe Welding

Describe the 3G plate position.

Answer

The plate stands upright and the weld runs up or down. This is the vertical position.

#464 · Level 2 Plate and Pipe Welding

Describe the 4G plate position.

Answer

The plate is above the welder and the weld is made from underneath. This is the overhead position.

#465 · Level 2 Plate and Pipe Welding

What is a groove angle (included angle)?

Answer

It is the total angle between the two prepared edges of the joint. Two 30° bevels give a 60° groove angle.

#466 · Level 2 Plate and Pipe Welding

What is a root opening?

Answer

It is the gap between the two pieces at the bottom of the joint. Its size comes from the WPS and is usually wider with backing than for open root.

#467 · Level 2 Plate and Pipe Welding

What is a root face (land)?

Answer

It is the small flat, unbeveled strip at the bottom of a prepared edge. It helps control penetration and burn through.

#468 · Level 2 Plate and Pipe Welding

What is the purpose of backing on a groove weld?

Answer

Backing is a bar tacked behind the joint to support the molten puddle. It makes burn through less likely and is easier to learn than open root.

#469 · Level 2 Plate and Pipe Welding

What is an open root joint?

Answer

An open root joint has no backing behind it. The welder must fuse both edges and form the back bead with nothing to support the puddle.

#470 · Level 2 Plate and Pipe Welding

What does a test done with backing qualify for open root joints?

Answer

A test done with backing does not qualify a welder for open root joints. A test done open root generally qualifies for backed joints, but confirm this in the code.

#471 · Level 2 Plate and Pipe Welding

What is a root pass?

Answer

It is the first pass in the joint. It fuses the root and sets the shape of the back side, and later passes cannot repair a bad one.

#472 · Level 2 Plate and Pipe Welding

What is the job of the hot pass?

Answer

The hot pass is run soon after the root pass at slightly higher current. It burns out slag traps and smooths the root bead edges for the next passes.

#473 · Level 2 Plate and Pipe Welding

What are fill passes and the cap pass?

Answer

Fill passes build the joint up to nearly flush with the plate surface. The cap pass is the last layer and sets the final weld width and reinforcement.

#474 · Level 2 Plate and Pipe Welding

What is reinforcement on a groove weld?

Answer

Reinforcement is extra weld metal standing above the plate surface. Codes limit it, a common limit being about 1/8 inch, but check the code or WPS.

#475 · Level 2 Plate and Pipe Welding

What is a keyhole in open root welding?

Answer

A keyhole is a small round opening at the leading edge of the puddle where the arc has melted through the root face. A steady keyhole shows full penetration.

#476 · Level 2 Plate and Pipe Welding

What does a growing keyhole mean?

Answer

A growing keyhole means the weld is about to burn through. The welder should speed up or lower the current.

#477 · Level 2 Plate and Pipe Welding

What does a closing keyhole mean?

Answer

A closing keyhole means penetration is being lost. Incomplete joint penetration will follow unless the welder slows down or adds heat.

#478 · Level 2 Plate and Pipe Welding

What are the 1G, 2G, 5G and 6G pipe positions?

Answer

1G is rolled horizontal pipe. 2G is vertical pipe with a horizontal weld. 5G is fixed horizontal pipe. 6G is fixed pipe at 45°.

#479 · Level 2 Plate and Pipe Welding

Why is the 5G pipe position difficult?

Answer

The pipe is fixed horizontal and cannot be turned. The welder works through flat, vertical and overhead positions in one weld.

#480 · Level 2 Plate and Pipe Welding

What does a 6G pipe test qualify?

Answer

In common AWS structural practice, 6G qualifies all positions on pipe and plate groove welding, within the thickness and diameter range tested. Confirm against your code.

#481 · Level 2 Plate and Pipe Welding

What does a 3G and 4G plate test qualify?

Answer

In common AWS structural practice, passing 3G and 4G on plate qualifies all positions of groove and fillet welding on plate. Confirm against your code.

#482 · Level 2 Plate and Pipe Welding

Does a SMAW test qualify a welder for GMAW?

Answer

No. Changing the welding process requires a new test.

#483 · Level 2 Plate and Pipe Welding

What is the E6010 electrode used for?

Answer

E6010 is a deep penetrating electrode run on direct current electrode positive. It is a standard choice for open root passes on pipe.

#484 · Level 2 Plate and Pipe Welding

What is the E7018 electrode used for?

Answer

E7018 is a low hydrogen electrode used for fill and cap passes. It gives smooth, strong, ductile welds.

#485 · Level 2 Plate and Pipe Welding

Why is E7018 stored in a heated rod oven?

Answer

Low hydrogen electrodes absorb moisture. Moisture puts hydrogen into the weld and can cause cracking, so oven temperature and time limits must be followed.

#486 · Level 2 Plate and Pipe Welding

What does the 70 in E7018 mean?

Answer

It means the weld metal has a tensile strength of 70 ksi, or 70,000 pounds per square inch.

#487 · Level 2 Plate and Pipe Welding

What does the T in E71T-1 mean?

Answer

T means tubular, so the wire is a flux cored wire. E71T-1 is a gas shielded wire that works in all positions.

#488 · Level 2 Plate and Pipe Welding

What shielding gas is common for GMAW on steel?

Answer

A mix of 75 percent argon and 25 percent carbon dioxide is common. Always follow the WPS.

#489 · Level 2 Plate and Pipe Welding

Which GMAW transfer mode suits out of position welding?

Answer

Short circuit transfer suits out of position welding and thin metal because it runs at low heat. Spray transfer works only in flat and horizontal.

#490 · Level 2 Plate and Pipe Welding

What is the difference between gas shielded and self shielded FCAW?

Answer

Gas shielded flux cored wire needs outside shielding gas. Self shielded wire makes its own shielding from the flux in the core.

#491 · Level 2 Plate and Pipe Welding

What shielding gas does GTAW use?

Answer

GTAW uses argon. The tungsten electrode does not melt and the filler rod is added by hand.

#492 · Level 2 Plate and Pipe Welding

What is a purge in pipe welding?

Answer

A purge is argon fed inside the pipe so the back of the root weld sits in inert gas. It is used on metals such as stainless steel.

#493 · Level 2 Plate and Pipe Welding

What is sugaring?

Answer

Sugaring is a gray, crusty oxidized surface on the back of a root weld caused by a poor purge.

#494 · Level 2 Plate and Pipe Welding

What is a guided bend test?

Answer

A strip cut from the weld is bent into a U shape around a plunger. The convex outer surface is then checked for open discontinuities.

#495 · Level 2 Plate and Pipe Welding

What is the difference between a face bend, root bend and side bend?

Answer

A face bend puts the weld face on the outside of the curve and a root bend puts the root there. A side bend stretches the full weld thickness and is used on thicker material.

#496 · Level 2 Plate and Pipe Welding

What plunger diameter is common for a 3/8 inch bend specimen?

Answer

A common rule is 4 times the thickness, so 4 × 3/8 = 1.5 inches. Confirm in the code.

#497 · Level 2 Plate and Pipe Welding

What is a common bend test limit on a single discontinuity?

Answer

In common AWS structural practice, no single discontinuity on the convex surface may exceed 1/8 inch in any direction. Confirm in your code.

#498 · Level 2 Plate and Pipe Welding

What is a common limit on the sum of discontinuities in a bend test?

Answer

In common AWS structural practice, the sum of discontinuities over 1/32 inch and up to 1/8 inch may not exceed 3/8 inch. Confirm in your code.

#499 · Level 2 Plate and Pipe Welding

How is heat input calculated?

Answer

Heat input in kJ per inch equals volts × amps × 60 ÷ (travel speed in inches per minute × 1,000). At 22 V, 100 A and 8 in/min it is 16.5 kJ/in.

#500 · Level 2 Plate and Pipe Welding

What is the SENSE Level 2 module structure?

Answer

Level 2 has five compulsory modules plus optional plate, pipe and fabrication modules. The knowledge exam pass mark is 75 percent, except Level 1 safety which needs 100 percent.